

Roll No.

Total No. of Questions : 9]
(1049)

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**B.C.A. (CBCS) RUSA IInd Semester
Examination**

4387

DIGITAL ELECTRONICS

Paper : BCA-0203

Time : 3 Hours]

[Maximum Marks : 70

Note :- Attempt *five* questions in all, selecting one question each from Unit-I to Unit-IV. Part-A (Q. No. 1) is compulsory.

Part-A

(Compulsory Question)

1. (A) Attempt all parts. Select the correct option for MCQ's.

(i) The output of an AND gate with 3-inputs A, B and C is HIGH when :

(a) $A = 1, B = 1, C = 0$

CH-713

(1)

Turn Over

(b) $A = 0, B = 0, C = 0$

(c) $A = 1, B = 0, C = 0$

(d) $A = 1, B = 1, C = 1$

(ii) When used with an IC, what does the term 'QUAD' indicate ?

(a) 2 circuits

(b) 4 circuits

(c) 8 circuits

(d) 6 circuits

(iii) The format used to present the logic output for the various combinations of logic inputs to a gate is called a (an) :

(a) Boolean Constant

(b) Boolean Variable

(c) Truth Table

(d) Input Logic Function

(iv) Which of the following expressions is in the sum-of-products (SOP) form ?

(a) $(A + B)(C + D)$

(b) $(A) B (CD)$

(c) $AB + CD$

(d) $AB (CD)$

(v) The commutative law of Boolean addition states that $A + B = A \times B$. (True/False)

(vi) The Boolean expression $C + CD$ is equal to

(vii) When transistors are used in digital circuits they usually operate in the :

(a) active region

(b) breakdown region

~~(c)~~ saturation and cutoff regions

(d) linear region

(viii) On the Master-Slave flip-flop, when it is master enabled ?

- (a) When the gate is HIGH
- (b) When the gate is LOW
- (c) Both of these
- (d) None of these



(ix) Under normal conditions a diode conducts current when it is :

- (a) Reverse biased
- (b) Forward biased
- (c) Saturated
- (d) Avalanched

(x) An n -type semiconductor material :

- (a) is intrinsic
- (b) has trivalent impurity atoms added
- (c) has pentavalent impurity atoms added
- (d) requires no doping

$$1 \times 10 = 10$$

(B) Answer the following in 25 to 50 words :

- (i) State the associative property of Boolean Algebra.
- (ii) What is meant by Karnaugh map method ?
- (iii) State advantages and disadvantages of TTL.
- (iv) What is a Decoder ?
- (v) Define Minterm and Maxterm. 4×5=20

Part-B

(Unit-I)

- 2. (a) Discuss the working of $p-n$ junction diode.
- (b) Explain energy bands in solids. 5,5
- 3. (a) Discuss Saturated and Non-saturated Logic.
- (b) Which is faster ECL or TTL ? Explain. 5,5

Part-C

(Unit-II)

- 4. (a) Simplify the expressions using Boolean Algebra :
 - (i) $A\bar{B}C + ABC$
 - (ii) $(\bar{A} + B + C)(A + B + \bar{C})$

(b) Give the circuit diagram of XOR gate. Also give its truth table. 4,6

5. (a) How can you connect NAND gates to get an OR gate ?

(b) What are the two basic rules used to draw equivalent gates ? 6,4

Part-D

(Unit-III)

6. (a) Simplify the following function in sum-of-product SOP form using four variable Karnaugh's map :

$$F(A, B, C, D) = \sum m(0, 1, 2, 3, 4, 5, 7, 11, 15)$$

(b) What are redundant groups in K-map ? 8,2

7. (a) Explain how basic gates can be realized using NAND gates. Also give the diagram.

(b) What do you mean by Combinational Circuit ? 6,4

Part-E

(Unit-IV)

8. (a) What is a Multiplexer ? Explain difference between MUX and DEMUX.
- (b) What do you mean by Shift-Registers ? Discuss. 5,5
9. (a) Explain the working and circuit of a Half-Adder.
- (b) Give the design of 3×8 decoder. 6,4

